

Shock Formation and Weak Solutions for Burgers' Equation

Equation and Setup

We consider Burgers' equation (all coefficients equal to 1):

$$u_t + uu_x = 0$$

with initial condition:

$$u(x, 0) = u_0(x)$$

This is a nonlinear first-order conservation law.

Method of Characteristics

Rewrite as:

$$u_t + a(u)u_x = 0, \quad a(u) = u$$

Characteristic equations:

$$\frac{dx}{dt} = u \quad \frac{du}{dt} = 0$$

Characteristic Solution

From $\frac{du}{dt} = 0$:

$$u = u_0(x_0)$$

From $\frac{dx}{dt} = u$:

$$x(t) = x_0 + u_0(x_0)t$$

Implicit solution:

$$u(x, t) = u_0(x_0), \quad x = x_0 + u_0(x_0)t$$

Gradient Evolution

Differentiate implicitly:

$$x = x_0 + u_0(x_0)t$$

Then:

$$\frac{\partial x}{\partial x_0} = 1 + u'_0(x_0)t$$

Thus:

$$u_x = \frac{u'_0(x_0)}{1 + u'_0(x_0)t}$$

Shock Formation

Blow-up occurs when:

$$1 + u'_0(x_0)t = 0$$

So earliest shock time:

$$t_s = -\frac{1}{\min u'_0(x)}$$

provided $\min u'_0(x) < 0$.

Characteristics intersect $\Rightarrow$ multi-valued solution.

Geometric Interpretation

- ▶ Characteristics are straight lines
- ▶ Speed depends on initial data
- ▶ Faster characteristics overtake slower ones
- ▶ Leads to steepening and shock formation

Breakdown of Classical Solutions

After t_s :

- ▶ Solution becomes multi-valued
- ▶ Classical derivative u_x blows up
- ▶ Need generalized (weak) solutions

Why "Conservation Law"?

A PDE of the form

$$u_t + (f(u))_x = 0$$

expresses conservation of a quantity u .

Interpretation:

- ▶ $u(x, t)$ = density of a conserved quantity
- ▶ $f(u)$ = flux (rate of transport)

The equation states: change in density = net flux in/out.

Integral Conservation Statement

For any interval $[a, b]$:

$$\frac{d}{dt} \int_a^b u(x, t), dx = f(u(a, t)) - f(u(b, t))$$

Meaning:

- ▶ Total amount changes only through boundary flux
- ▶ No creation or destruction inside the interval

Differentiating this relation formally leads back to:

$$u_t + (f(u))_x = 0$$

Conservation Form

Write equation as:

$$u_t + \left(\frac{u^2}{2} \right)_x = 0$$

This is a conservation law with flux:

$$f(u) = \frac{u^2}{2}$$

Weak Formulation

Multiply by test function $\phi(x, t)$:

$$\int \int \left(u \phi_t + \frac{u^2}{2} \phi_x \right) dx dt = 0$$

(after integration by parts)

Defines weak solutions allowing discontinuities.

Origin of Rankine–Hugoniot Condition

We derive the jump condition from conservation over a moving control volume.

Consider an interval $[a(t), b(t)]$ containing the shock at $x = s(t)$.

Conservation law:

$$\frac{d}{dt} \int_{a(t)}^{b(t)} u(x, t), dx = - [f(u(b(t), t)) - f(u(a(t), t))]$$

Split the integral at the shock and account for motion of $s(t)$.

Jump Condition Derivation

Write:

$$\frac{d}{dt} \left(\int_{a(t)}^{s(t)} u_L, dx + \int_{s(t)}^{b(t)} u_R, dx \right)$$

Differentiate using Leibniz rule:

$$s'(t)(u_L - u_R) = f(u_L) - f(u_R)$$

Thus:

$$s'(t) = \frac{f(u_R) - f(u_L)}{u_R - u_L}$$

Rankine–Hugoniot from Weak Formulation

Start from the weak form:

$$\int \int (u \phi_t + f(u) \phi_x), dx, dt = 0$$

Choose a test function ϕ supported in a small neighborhood of the shock curve $x = s(t)$.

Split the domain into left/right regions and integrate by parts on each side.

Boundary terms along the shock give the jump condition.

Computation of Shock Contribution

The boundary contribution along $x = s(t)$ yields:

$$\int (-s'(t)[u] + [f(u)])\phi(s(t), t), dt = 0$$

where $[u] = u_R - u_L$ and $[f(u)] = f(u_R) - f(u_L)$.

Since this holds for all test functions ϕ :

$$-s'(t)[u] + [f(u)] = 0$$

Thus:

$$s'(t) = \frac{f(u_R) - f(u_L)}{u_R - u_L}$$

Shock Conditions

Assume a discontinuity moving at $x = s(t)$.

Rankine–Hugoniot condition:

$$s'(t) = \frac{f(u_R) - f(u_L)}{u_R - u_L}$$

For Burgers:

$$s'(t) = \frac{u_R^2/2 - u_L^2/2}{u_R - u_L} = \frac{u_L + u_R}{2}$$

Entropy Condition

Not all weak solutions are physical.

Entropy condition (Lax):

$$u_L > u_R$$

Characteristics must enter the shock from both sides.

Ensures uniqueness and physical relevance.

Shock vs Rarefaction

- ▶ If $u_L > u_R$: shock forms
- ▶ If $u_L < u_R$: rarefaction wave

Rarefaction solution is continuous and self-similar.

Summary

- ▶ Characteristics provide explicit solution until crossing
- ▶ Gradient blow-up leads to shocks
- ▶ Weak solutions extend beyond breakdown
- ▶ Rankine–Hugoniot gives shock speed
- ▶ Entropy condition selects physical solution

Vanishing Viscosity Idea

Select physically relevant sols with the viscous Burgers eq:

$$u_t + uu_x = \varepsilon u_{xx}$$

with $\varepsilon > 0$ small.

- ▶ For each $\varepsilon > 0$, solution remains smooth
- ▶ As $\varepsilon \rightarrow 0$, solutions develop sharp layers
- ▶ Limit defines the **viscosity solution**

Shock Selection via Viscosity

Key result:

- ▶ The limit $\varepsilon \rightarrow 0$ selects the entropy solution
- ▶ Discontinuities with $u_L > u_R$ persist
- ▶ Discontinuities with $u_L < u_R$ smooth into rarefactions

Heuristic picture:

- ▶ Diffusion resolves discontinuity into a thin layer
- ▶ Balance between convection and diffusion determines profile
- ▶ In the limit, only admissible shocks remain

Traveling Wave Profile

Seek steady profile in moving frame $\xi = x - st$:

$$-su' + uu' = \varepsilon u''$$

Integrate once:

$$\varepsilon u' = \frac{u^2}{2} - su + C$$

Matching u_L, u_R gives Rankine–Hugoniot condition for s .

Monotone profile exists only if $u_L > u_R$.